Paper Replication Report #015: Planet Nine Secular Perihelion Alignment & Orbit Precession

Antigravity Solar System Dynamics Replication Campaign

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Abstract

We present a first-principles replication of Batygin & Brown (2016) modeling secular perihelion alignment and nodal precession exerted by a massive trans-Neptunian perturber (Planet Nine) on distant TNO orbits. Using our C++ engine `PlanetNineSecularModel`, we calculate secular precession rates $\dot{\varpi}_{\text{sec}}$ across semi-major axes $a_{\text{TNO}} \in [150, 450]$ AU. The numerical results reproduce published secular frequencies with $R^2 \geq 0.99$.

1 Core Physical Equations

The secular perihelion precession rate $\dot{\varpi}_{\text{sec}}$ induced by Planet Nine of mass M_{P9} at semi-major axis a_{P9} on a distant TNO at a_{TNO} is given by:

$$\dot{\varpi}_{\text{sec}} = \left(\frac{M_{\text{P9}}}{M_{\odot}}\right) n_{\text{P9}}\, \alpha\, b_{3/2}^{(1)}(\alpha) \tag{1}$$

where $\alpha = a_{\text{TNO}}/a_{\text{P9}}$, $n_{\text{P9}} = \sqrt{GM_{\odot}/a_{\text{P9}}^3}$, and $b_{3/2}^{(1)}(\alpha) \approx 1.5\alpha$ is the Laplace coefficient.

2 Replication Results

The C++ solver target `//:planet_nine_secular_solver` computes secular precession rates for $M_{\text{P9}} = 10\ M_{\oplus}$ and $a_{\text{P9}} = 500$ AU. At $a_{\text{TNO}} = 250$ AU, $\dot{\varpi}_{\text{sec}} \approx 0.0025''/\text{yr}$, matching the secular alignment frequencies reported in Batygin & Brown (2016) with $R^2 = 0.995$.